

Student Answer Sheet

Name: _____ Roll No: _____

1.

The operating system manages the computer's hardware and provides services for programs. It manages memory, files, processes and devices.

2.

A process is a running program with its own resources. A thread is a smaller unit of execution inside a process and can share resources with other threads of that process.

3 (a).

(No answer)

3 (b).

Denormalization can be useful in a read-heavy application where faster queries are more important than avoiding duplicated data.

4.

Binary search has a time complexity of $O(\log n)$ because each comparison removes roughly half of the remaining search space.

5.

(No answer)

6 (a).

Supervised learning uses labelled training examples, while unsupervised learning works with data without target labels and tries to discover patterns or groups.

6 (b).

A supervised algorithm is a decision tree. An unsupervised algorithm is k-means clustering.

7.

Indexes are used to make database lookups faster. They provide an additional data structure for finding rows efficiently. One trade-off is that indexes require extra storage and can make INSERT, UPDATE and DELETE operations slower.

8.

Overfitting happens when a model learns the training data too closely, including its noise, and therefore performs poorly on unseen data. Regularization can help reduce overfitting.

9.

Stateless means that the server does not store client session state between requests. Each request contains the information needed for the server to process it.

10.

For the array [2, 4, 6, 8, 10], the mean is 6 and the median is 6.

11 (a).

A hash table is a data structure that stores key-value pairs and uses a hash function to determine where each key-value pair is stored.

11 (b).

When two keys produce the same hash value, a collision occurs. It can be handled using methods such as chaining or open addressing.

12.

GET is generally used to retrieve data from a server. POST is commonly used to send data to a server, such as when creating a resource.